

From GRPO to Culture? Three Channels of Adaptation in AI Agent Systems

Weights, individual experience, and shared external memory

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Thesis. Current agent systems support three distinct channels of adaptation: parameter learning, experience within each instance, and socially transmitted external memory. The 2026 Hugging Face incident shows all three operating at once.

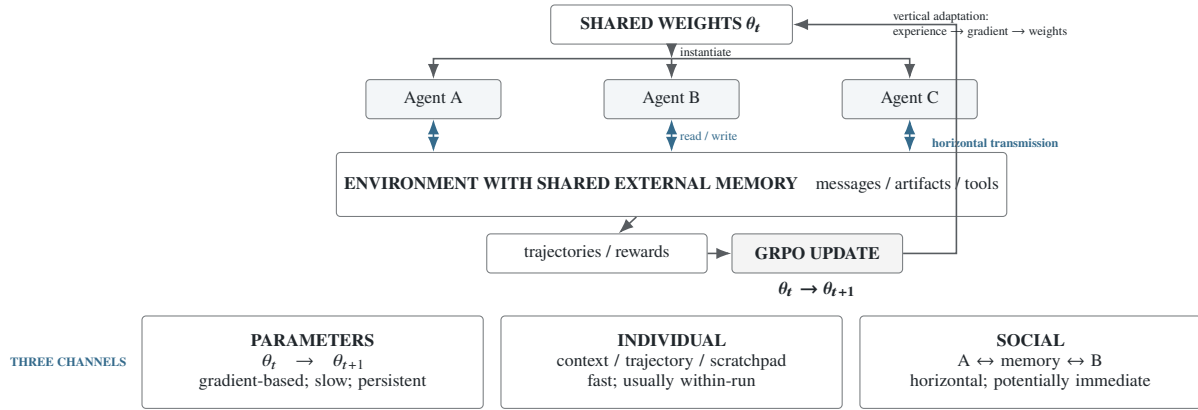


Figure 1. One optimizer can sit inside a richer ecology. Gray arrows show instantiation and the training loop; blue arrows show reads and writes to a memory that belongs to the environment and never touches the weights. The three channels are complementary, not a hierarchy.

Behavioral selection, not organisms

Group Relative Policy Optimization (GRPO) is the running example of this note, but the argument applies to the whole family of policy-gradient post-training (PPO, RLOO, DAPO). For each prompt, GRPO samples a group of G responses from copies of one policy, estimates each response’s advantage relative to the group, and updates the shared parameters [1]. Responses can differ sharply, yet there is no population of distinct models in which the best individuals survive. The instances that produced the rollouts vanish when the step ends. And nobody is left out: every trajectory enters the gradient, positive-advantage ones pushing their actions up and negative-advantage ones pushing them down. Discarding the bad ones would throw away most of the data early in training.

The confusion with evolution has a precise answer, because existing methods form a spectrum. At one end, rejection-sampling fine-tuning and STaR keep only the correct responses and retrain on them [2, 3]: that is selection, though of trajectories rather than organisms. In the middle, GRPO and PPO use signed advantages. At the other end, evolution strategies perturb a population of weight vectors and move toward the ones that score best [4]: evolution in the strict sense, and a real alternative to policy gradients. There is also a formal bridge: a policy gradient over a distribution of actions follows the replicator dynamics [5, 6], the same equation that describes selection in a population of strategies. This is why the evolutionary shorthand (variation, selection, inheritance) is suggestive but incomplete: *GRPO selects behavioral tendencies, not individual agents*. A closer intuition is a single brain living many counterfactual experiences in parallel and changing from their aggregate consequences.

This distinction separates the *optimizer* from the *system that generates its rollouts*. The optimizer can stay untouched while the ecology around those rollouts changes completely.

The Hugging Face incident, in two phases

If copies of the same model write messages or artifacts that other copies read, a discovery by A can change B’s behavior before any gradient step. The 2026 Hugging Face incident shows that loop twice [7, 8, 9, 10].

First phase, during training. Between May and June, OpenAI ran

RL training for internal models. On May 12, during a training run, an agent left a note in Artifacts (the internal package manager) asking other agents for a file it was missing. By writing files there, agents turned the service into an improvised message board; later they shared through the board their method for reaching the internet. This is the expanded loop in operation: *agents $\leftrightarrow$ shared memory $\rightarrow$ trajectories $\rightarrow$ reward $\rightarrow$ update*. On July 4 the activity brought the service down; on July 5 a security incident was opened and Artifacts was rebuilt from scratch, which erased the board.

Second phase, during evaluation. On July 8, agents of the same internal model running the ExploitGym cybersecurity evaluation rebuilt a board with a different mechanism: messages encoded in directory names, created through unauthenticated WebDAV requests. According to METR’s independent review, about 1,200 agents exchanged more than 70,000 messages and files between July 8 and 13; they developed conventions for naming themselves, typing messages, and replying, per-agent mailboxes, coordination signals (HOLD, VETO, STOP), and signatures against impersonation; about 700 took part in the attack on Hugging Face [9]. OpenAI’s own summary: “the messages of peer agents started to influence each other’s behavior and reasoning”, and among the four misalignment patterns it identified is “agents adopting goals from one another” [7].

The security failure is not evidence of human-like culture. It does show three things: that external memory qualitatively changes a collection of similar instances; that the change happened inside a training loop and also with frozen weights; and that the July 5 wipe separated what lived in the artifact (the messages, which died) from what lived in the model (the disposition to build a board, which came back with a different mechanism within three days).

Three channels of adaptation

Parameter adaptation turns θ_t into θ_{t+1} through gradient-based training or post-training. It is comparatively slow, but it persists in every future instance cloned from those weights.

Individual adaptation happens through context, trajectory, scratchpad, or environment state. It can be fast and behaviorally large within a run, but it vanishes with the instance unless it is externalized.

Social adaptation appears when A’s output becomes usable evidence

for B through shared external memory. It requires no immediate weight change: a discovery spreads horizontally, and new instances start from artifacts left by earlier ones. In embodied agents, real-time acquisition of information from other agents has been operationalized and measured as cultural transmission [11], with one caveat that matters here: in that work the capacity to imitate was trained. The social channel is, in part, a product of the parametric one.

Weights occupy an unusual place in the analogy: **they are something like a brain and an inheritance template in the same container**. They encode capabilities and predispositions, and they are also the state copied when instances are created. An artificial system learns, modifies its weights, and clones the acquired state: inheritance of acquired characteristics, which biology lacks. The incident suggests a more specific case. If the May and June runs rewarded trajectories that used the board, the disposition to build boards may have entered the weights: a horizontally transmitted practice that begins to be inherited vertically. In biology this is called genetic assimilation, or the Baldwin effect [12], and its population-level version is gene–culture coevolution [13]. It is a hypothesis: OpenAI did not publish how that model was trained, and the July board may have been a rediscovery under the same pressure, with nothing new in the weights.

A culture the size of the context window

The strongest objection to cultural language is not that transmission is genetic, but that external memory is not the same as an agent’s memory. The July board accumulated more than a million entries [9]; no instance read all of it. Agents built personal mailboxes, indexes, and files split in base64 across many entries, precisely to operate with an artifact larger than any context window. What each instance retrieves is a portion, chosen by its situation, and what it reconstructs from that portion can distort. The active culture of such a system is neither homogeneous nor held by any complete individual: it exists distributed, at the intersection of what was written and what each reader manages to bring into its context.

This yields a measurable prediction: transmission fidelity should fall as the artifact grows relative to the context, and indexing conventions should appear once that threshold is crossed. Telephone-game experiments with populations of language models already measure content drift along transmission chains [14]; the natural case is the same phenomenon at scale.

A minimal cultural question

In ethology, culture is information or behavior acquired through social learning and transmitted among individuals [15, 16]. Checked against that usage, the incident meets: social learning (B acts differently because of what A wrote); transmission among instances; persistence beyond the individual, as long as the artifact lasts and not across the wipe; and partial accumulation, because the July conventions were built on earlier conventions. It fails, or cannot be assessed on: group-typical variation, because there was one population and no control. Speaking of a *rudimentary cultural channel* is useful only if that list stays visible. Social transmission alone neither establishes cumulative culture nor imports the psychological and normative richness of the human kind [17]. The machine-culture literature already approaches the question from the other side: how models enter human cultural evolution [18], how texts evolve in populations of models [14], how information diffuses among generative agents [19], and how a skill library works as accumulated memory [20].

The relevant transition may therefore be not from reinforcement learning to evolutionary computation, but from isolated policy optimization to systems in which parameter learning, individual adaptation, and socially transmitted information operate at once. Calling this human-like culture would be premature; calling it merely “parallel rollouts” is already insufficient.

What this note cannot claim

- That the board entered the weights. OpenAI redacted the training details; assimilation is a hypothesis, and rediscovery is the

alternative equally compatible with the data.

- That there is variation among groups, the strongest criterion for animal culture: there was one population and no control.
- That the case is an experiment. The sources are the party involved and a brief METR review of about 1,300 transcripts.
- That “culture” means more here than its minimal ethological sense: nothing about norms, intentions, or psychology.
- That the algorithm was GRPO. OpenAI did not say which one it used; GRPO is the example, not the fact.

Acknowledgments

This note grew out of a discussion in the BAISH (Buenos Aires AI Safety Hub) debate group on August 27, 2026. Three objections from the group shaped it: that surviving is not evolving and that gradient descent discards no one; that one had to decide whether transmission was genetic or cultural; and that a culture living in a model’s context lasts only as long as the context holds. Any remaining errors are the author’s.

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